Algorithms

Lecture 6
Karatsuba for Multiplication (Divide and Conquer)

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Naive Algorithm

```
   12
x  34
-----
   48
  360
```

2*4=8
1*4=4
2*3=6
1*3=3

4 multiplications required

Naive Algorithm

```
X = 1234
Y = 5678

1234 x 5678
```

Divide and Conquer Approach

$X = X_l * 10^{n/2} + X_r$ [X_l and X_r contain leftmost and rightmost n/2 bits of X]

$Y = Y_l * 10^{n/2} + Y_r$ [Y_l and Y_r contain leftmost and rightmost n/2 bits of Y]

$$XY = (X_l * 10^{n/2} + X_r)(Y_l * 10^{n/2} + Y_r)$$
$$= 10^n X_l Y_l + 10^{n/2}(X_l Y_r + X_r Y_l) + X_r Y_r$$

$X = 1234$

$Y = 5678$

$X_l = 12$, $X_r = 34$

$Y_l = 56$, $Y_r = 78$

Divide and Conquer Approach

$X = X_l * 10^{n/2} + X_r$ [X_l and X_r contain leftmost and rightmost n/2 bits of X]

$Y = Y_l * 10^{n/2} + Y_r$ [Y_l and Y_r contain leftmost and rightmost n/2 bits of Y]

$$XY = (X_l * 10^{n/2} + X_r)(Y_l * 10^{n/2} + Y_r)$$
$$= 10^n \boxed{X_l Y_l} + 10^{n/2} (\boxed{X_l Y_r} + \boxed{X_r Y_l}) + \boxed{X_r Y_r}$$

We have divided the entire problem to 4 subproblems

Still same 4 multiplications required, not an optimization

Divide and Conquer Approach

$X = X_l * 10^{n/2} + X_r$ [X_l and X_r contain leftmost and rightmost n/2 bits of X]

$Y = Y_l * 10^{n/2} + Y_r$ [Y_l and Y_r contain leftmost and rightmost n/2 bits of Y]

$$XY = (X_l * 10^{n/2} + X_r)(Y_l * 10^{n/2} + Y_r)$$
$$= 10^n X_l Y_l + 10^{n/2} (X_l Y_r + X_r Y_l) + X_r Y_r$$

Change this term to something better

Divide and Conquer Approach

Tricky expression for middle two terms:

$X_l Y_r + X_r Y_l =$

Divide and Conquer Approach

Tricky expression for middle two terms:

$$X_lY_r + X_rY_l = (X_l + X_r)(Y_l + Y_r) - X_lY_l - X_rY_r$$

We already calculated these two,
No extra multiplication needed for them

Divide and Conquer Approach

Tricky expression for middle two terms:

$$X_l \times Y_r + X_r \times Y_l = (X_l + X_r) \times (Y_l + Y_r) - X_l Y_l - X_r Y_r$$

Instead of these two multiplications

Only one new multiplication needed

Divide and Conquer Approach

So the final value of XY becomes

$$XY = 10^{n} X_l Y_l + 10^{n/2} ((X_l + X_r)(Y_l + Y_r) - X_l Y_l - X_r Y_r) + X_r Y_r$$

3 multiplications required instead of 4

$$12 \times 56$$
$$= (12 \times 56) \times 10^{4}$$
$$+ ((12 + 34) \times (56 + 78) - 12 \times 56 - 34 \times 78) \times 10^{2}$$
$$+ 34 \times 78$$

Divide and Conquer Approach

Analyzing time complexity

$$XY = 10^{n} X_l Y_l + 10^{n/2}(X_l Y_r + X_r Y_l) + X_r Y_r$$

$$T(n) = 4T(n/2) + \Theta(n)$$

$$=$$

Divide and Conquer Approach

Analyzing time complexity

$$XY = 10^{n} X_l Y_l + 10^{n/2}((X_l + X_r)(Y_l + Y_r) - X_l Y_l - X_r Y_r) + X_r Y_r$$

$$T(n) = 3T(n/2) + \Theta(n)$$

$$= \Theta(n^{\wedge}(\log_2 3)) \text{ (master th}^{r}\text{m)}$$

$$= \Theta(n^{1.59}) \text{ which is less than } \Theta(n^2)$$

Divide and Conquer Approach

```
def karatsuba(x, y):
    if x < 10 or y < 10:
        return x * y

    n = max(len(str(x)), len(str(y)))
    m = ceil(n / 2)

    p = 10 ** m

    xh = x // p
    xl = x % p

    yh = y // p
    yl = y % p

    a = karatsuba(xh, yh)
    b = karatsuba(xl, yl)
    c = karatsuba(xh + xl, yh + yl)

    return a * (10 ** (2*m)) + (c - a - b) * (10 ** m) + b
```

$$XY = 10^{n} X_l Y_l + 10^{n/2}((X_l + X_r)(Y_l + Y_r) - X_l Y_l - X_r Y_r) + X_r Y_r$$

Consider m = n/2